EXPERIMENT 1: BASIC CELLULAR COMMUNICATION SYSTEM USING FIGMA

AIM:

To design an interactive diagram illustrating the components of a basic cellular communication system using Figma.

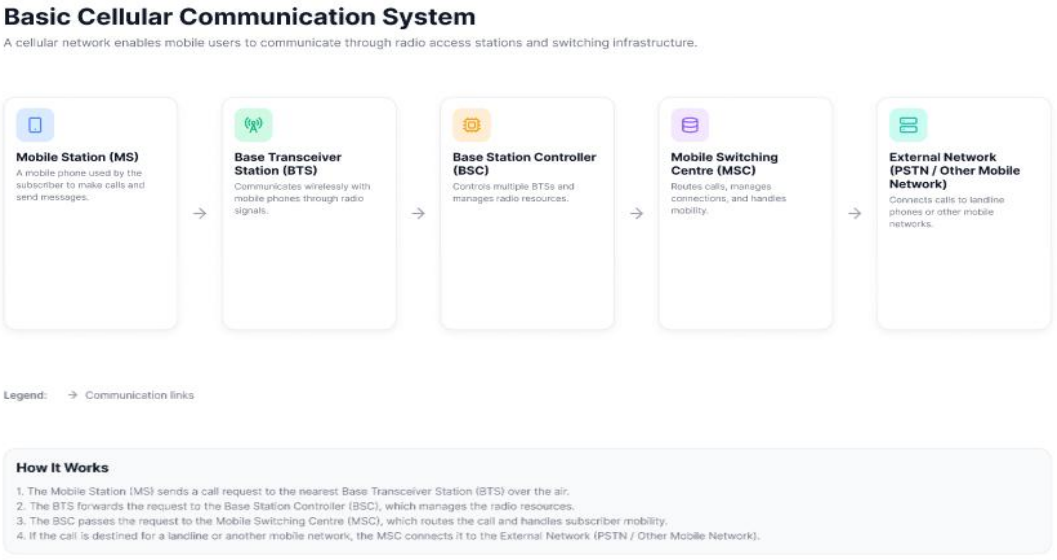
SOFTWARE REQUIRED:

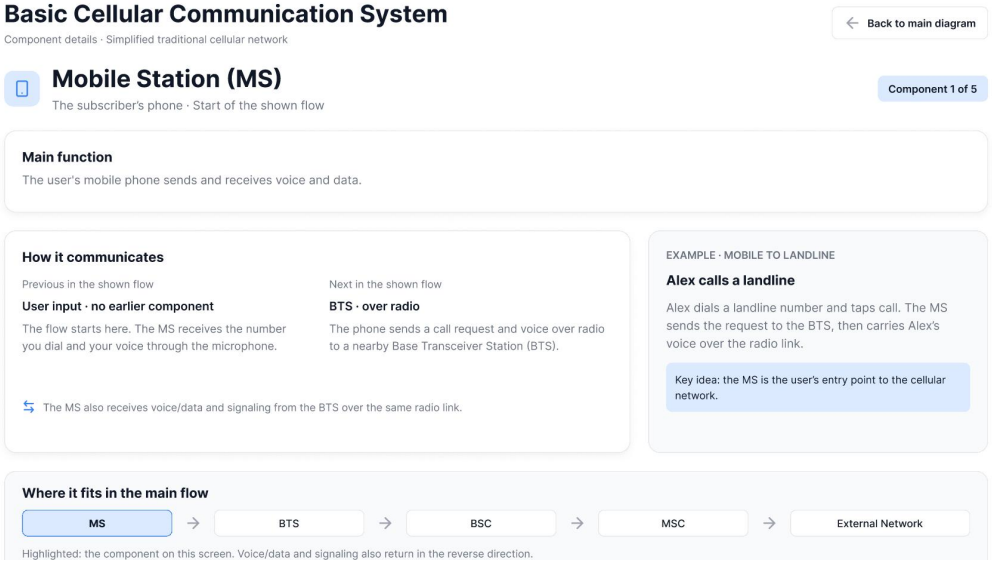
Figma (Web-based design tool).

PROCEDURE:

1. Open Figma and create a new design file.
2. Create a desktop frame of size 1440 × 900 pixels.
3. Add the title “Basic Cellular Communication System”.
4. Create cards for the five main components: Mobile Station (MS), Base Transceiver Station (BTS), Base Station Controller (BSC), Mobile Switching Centre (MSC), and External Network.
5. Add suitable icons, colors, and short descriptions for each component.
6. Connect the cards using directional arrows to represent communication flow.
7. Use the Prototype feature to connect the cards to their respective detail screens or overlays, with Back buttons to return to the main diagram.
8. Preview the design using the Present button and verify the interactions.

DESIGN


Basic Cellular Communication System
A cellular network enables mobile users to communicate through radio access stations and switching infrastructure.
Mobile Station (MS)
A mobile phone used by the subscriber to make calls and send messages.
Base Transceiver Station (BTS)
Communicates wirelessly with mobile phones through radio signals.
Base Station Controller (BSC)
Controls multiple BTSs and manages radio resources.
Mobile Switching Centre (MSC)
Routes calls, manages connections, and handles mobility.
External Network (PSTN / Other Mobile Network)
Connects calls to landline phones or other mobile networks.
Legend: Communication links
How It Works
1. The Mobile Station (MS) sends a call request to the nearest Base Transceiver Station (BTS) over the air.
2. The BTS forwards the request to the Base Station Controller (BSC), which manages the radio resources.
3. The BSC passes the request to the Mobile Switching Centre (MSC), which routes the call and handles subscriber mobility.
4. If the call is destined for a landline or another mobile network, the MSC connects it to the External Network (PSTN / Other Mobile Network).


Basic Cellular Communication System
Component details · Simplified traditional cellular network
Back to main diagram
Mobile Station (MS)
The subscriber's phone · Start of the shown flow
Component 1 of 5
Main function
The user's mobile phone sends and receives voice and data.
How it communicates
Previous in the shown flow
User input · no earlier component
The flow starts here. The MS receives the number you dial and your voice through the microphone.
Next in the shown flow
BTS · over radio
The phone sends a call request and voice over radio to a nearby Base Transceiver Station (BTS).
The MS also receives voice/data and signaling from the BTS over the same radio link.
EXAMPLE · MOBILE TO LANDLINE
Alex calls a landline
Alex dials a landline number and taps call. The MS sends the request to the BTS, then carries Alex's voice over the radio link.
Key idea: the MS is the user's entry point to the cellular network.
Where it fits in the main flow
MS
BTS
BSC
MSC
External Network
Highlighted: the component on this screen. Voice/data and signaling also return in the reverse direction.

Basic Cellular Communication System

Component details · Simplified traditional cellular network

Back to main diagram

Base Transceiver Station (BTS)

The radio access point · Between the phone and the controller

Component 2 of 5

Main function

Provides the radio connection between the mobile phone and the cellular network.

How it communicates

Previous in the shown flow

MS · over radio

The BTS receives the phone's radio signals, carrying the call request, voice/data and signaling.

Next in the shown flow

BSC · over backhaul

It forwards traffic to the Base Station Controller (BSC) over backhaul, the link into the network.

Traffic from the BSC travels back through the BTS and over radio to the MS.

EXAMPLE · MOBILE TO LANDLINE

Alex calls a landline

The BTS receives Alex's call request over the air and forwards it to the BSC. During the call, it relays voice between Alex's phone and the network.

Key idea: the BTS bridges the radio link and the rest of the network.

Where it fits in the main flow

MS → **BTS** → BSC → MSC → External Network

Highlighted: the component on this screen. Voice/data and signaling also return in the reverse direction.

Basic Cellular Communication System

Component details · Simplified traditional cellular network

Back to main diagram

Mobile Switching Centre (MSC)

The call switching point · Connects the cellular network to other networks

Component 4 of 5

Main function

Routes calls, manages connections, and coordinates mobility.

How it communicates

Previous in the shown flow

BSC · from the radio network

The MSC receives calls from the BSC side of the network and manages their connections and mobility.

Next in the shown flow

External Network · onward routing

It routes each call to the appropriate external destination: a landline or another mobile network.

The external network sends return traffic to the MSC, which passes it back toward the BSC.

EXAMPLE · MOBILE TO LANDLINE

Alex calls a landline

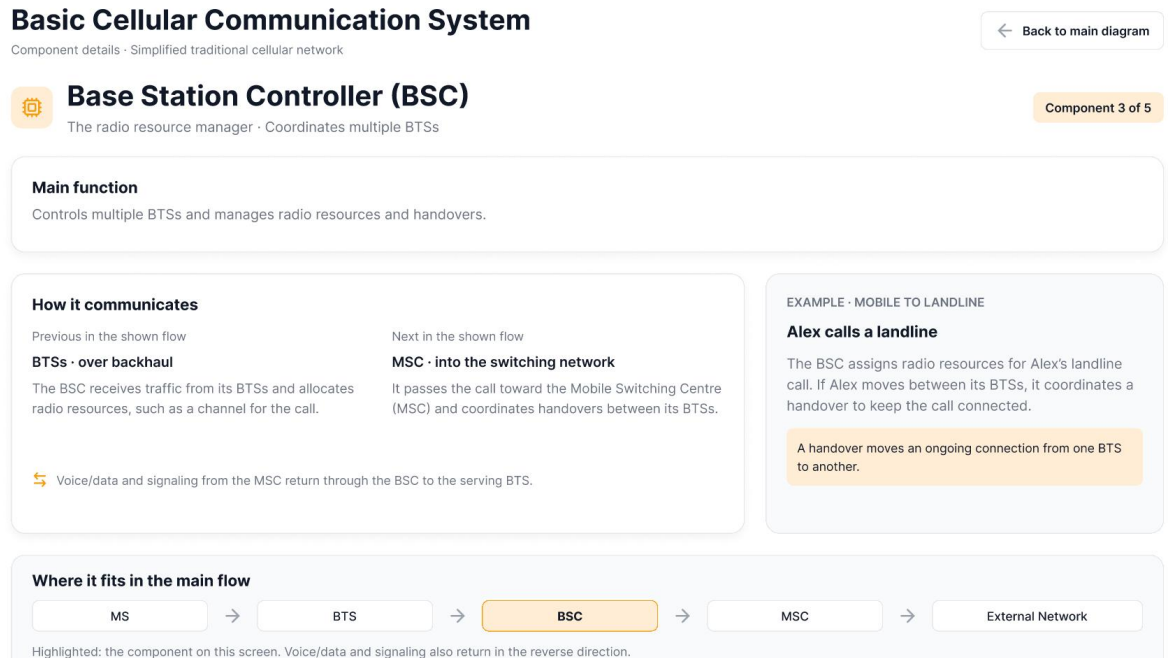
The MSC reads the destination for Alex's call and routes it to the landline network. It manages the connection and coordinates mobility while Alex moves.

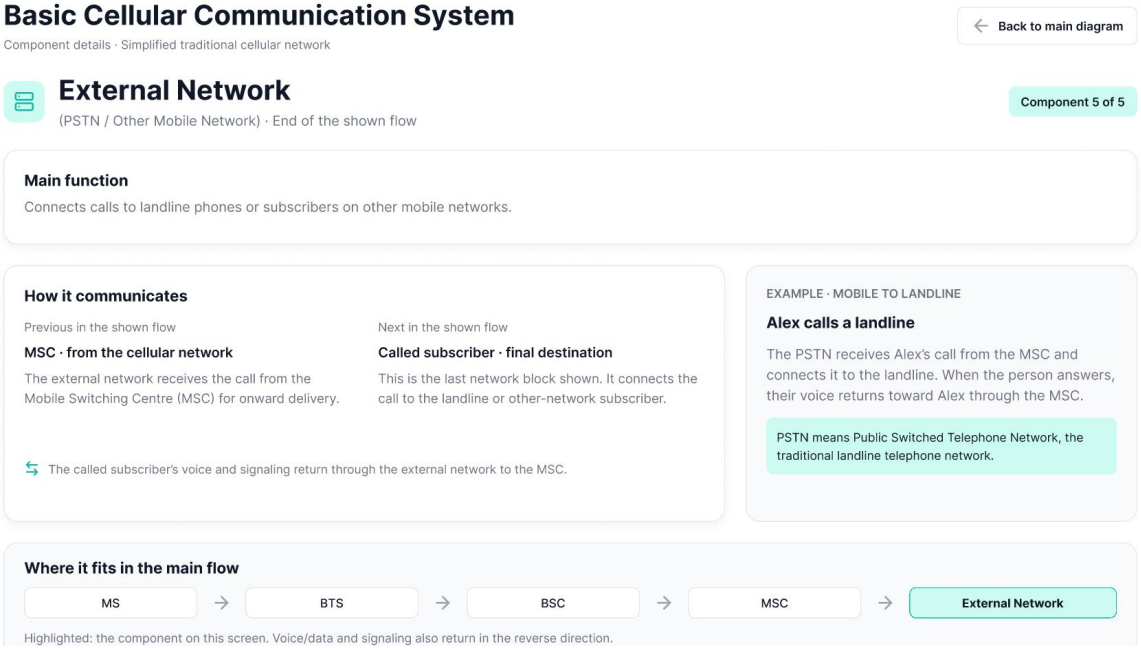
Key idea: the MSC chooses the onward route and manages the call connection.

Where it fits in the main flow

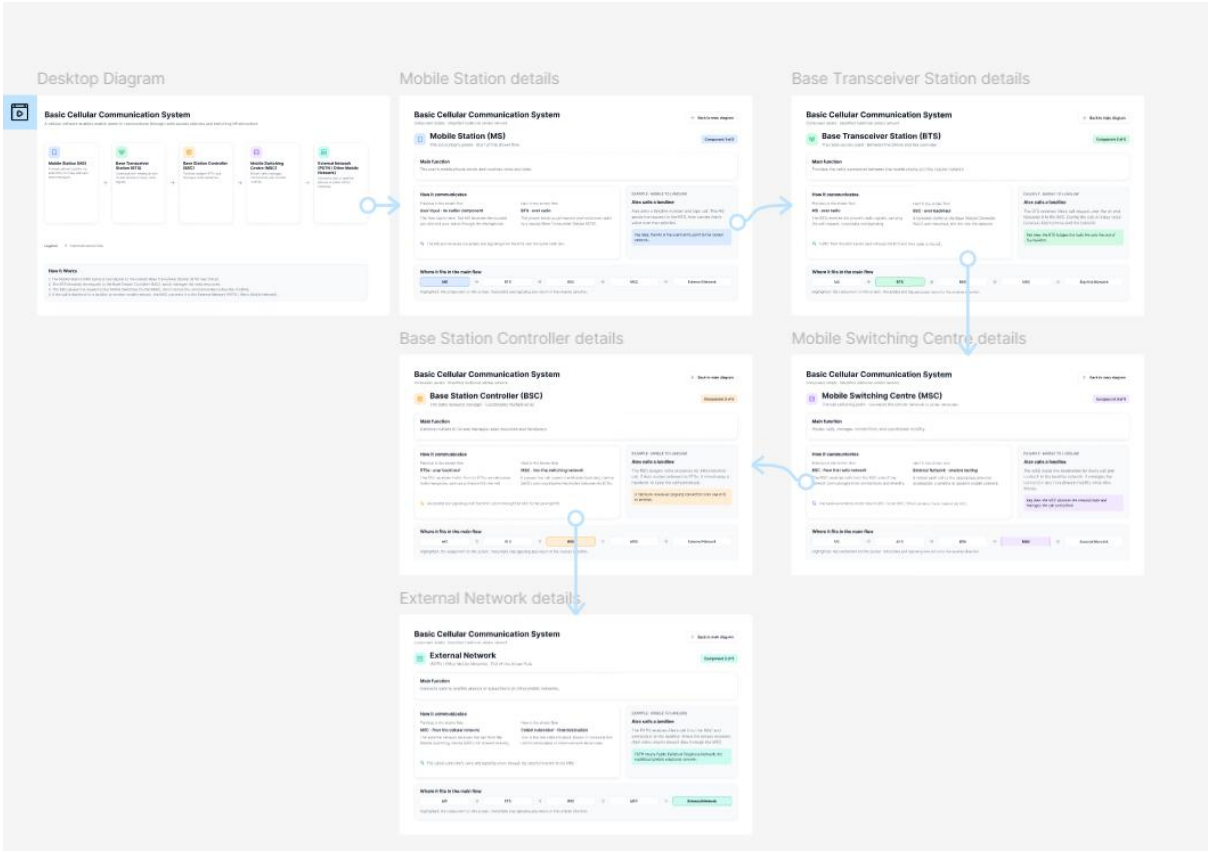
MS → BTS → BSC → **MSC** → External Network

Highlighted: the component on this screen. Voice/data and signaling also return in the reverse direction.


Basic Cellular Communication System
Component details · Simplified traditional cellular network
Back to main diagram
Base Station Controller (BSC)
The radio resource manager · Coordinates multiple BTSs
Component 3 of 5
Main function
Controls multiple BTSs and manages radio resources and handovers.
How it communicates
Previous in the shown flow
BTSs · over backhaul
The BSC receives traffic from its BTSs and allocates radio resources, such as a channel for the call.
Next in the shown flow
MSC · into the switching network
It passes the call toward the Mobile Switching Centre (MSC) and coordinates handovers between its BTSs.
Voice/data and signaling from the MSC return through the BSC to the serving BTS.
EXAMPLE · MOBILE TO LANDLINE
Alex calls a landline
The BSC assigns radio resources for Alex's landline call. If Alex moves between its BTSs, it coordinates a handover to keep the call connected.
A handover moves an ongoing connection from one BTS to another.
Where it fits in the main flow
MS
BTS
BSC
MSC
External Network
Highlighted: the component on this screen. Voice/data and signaling also return in the reverse direction.


Basic Cellular Communication System
Component details · Simplified traditional cellular network
Back to main diagram
External Network
(PSTN / Other Mobile Network) · End of the shown flow
Component 5 of 5
Main function
Connects calls to landline phones or subscribers on other mobile networks.
How it communicates
Previous in the shown flow
MSC · from the cellular network
The external network receives the call from the Mobile Switching Centre (MSC) for onward delivery.
Next in the shown flow
Called subscriber · final destination
This is the last network block shown. It connects the call to the landline or other-network subscriber.
The called subscriber's voice and signaling return through the external network to the MSC.
EXAMPLE · MOBILE TO LANDLINE
Alex calls a landline
The PSTN receives Alex's call from the MSC and connects it to the landline. When the person answers, their voice returns toward Alex through the MSC.
PSTN means Public Switched Telephone Network, the traditional landline telephone network.
Where it fits in the main flow
MS
BTS
BSC
MSC
External Network
Highlighted: the component on this screen. Voice/data and signaling also return in the reverse direction.

PROTOTYPE:



OUTPUT:

A cellular communication system diagram showing five major components connected in sequence, with descriptions and prototype interactions.

RESULT:

The basic cellular communication system diagram was designed in Figma, illustrating the components and their communication flow. Interactive navigation was planned using the Prototype feature.